DEEP RESEARCH / LEGAL INFORMATION RETRIEVAL

# V20 DIRECT-5

## A parallel breakthrough path for V16b under the strict 4B limit

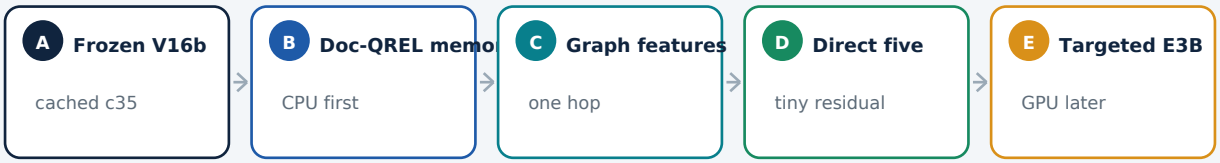
Decision memo | 30 August 2026

Current pipeline: V16b | Goal: improve the c35-to-five decision without disrupting the running V18 workload

**Decision: keep V18 E3B as a targeted evidence layer and launch a CPU-first candidate-conditioned direct top-five branch now.**

| 0.9210 | 0.967667 | 0.9600 | 82.9% |
|---|---|---|---|
| V16b public | locked c35 oracle | stretch target | headroom capture needed |

### CORE PATH




Promotion is sequential; A-D can run without loading CE/BQB. Stage E starts only after CPU gates pass.

The 0.96 target requires near-oracle behavior: approximately 83% of the locked c35-to-V16b headroom must be captured. This report therefore uses predeclared policy oracles, cross-fit leakage controls, paired bootstrap gates, and a GPU escalation only after CPU evidence passes.

# Executive decision

Continue V18 E3B, but change its role. Treat E3B as an evidence-feature experiment, not as the final selection architecture. In parallel, launch a CPU-first branch called V20 DIRECT-5 that learns the c35-to-five decision directly.

The highest-value new signal is candidate-conditioned QREL memory: for every candidate document in c35, retrieve the fit-fold training questions for which that document was relevant, and measure how well the current query matches those document-specific question prototypes. This asks a sharper question than V16b's global query-neighbor propagation: does this candidate historically answer questions like this one?

Add a small set of corpus-derived graph features only after the prototype branch proves useful. Do not begin with a GNN or a graph database. LegalIR research supports structural signal, but independent CaseLink reproduction found materially lower results than originally reported and no benefit from a more complex heterogeneous graph [R4]. The right risk level here is one-hop, degree-normalized features with strict ablations.

Train a tiny bounded-residual scorer on CPU and compare three decision policies: slot 5 only, ranks 4-5, and a direct five from a rank-3-to-12 ambiguity band. Only allow a full c35-to-five selector if its policy oracle and out-of-distribution tests justify that freedom.

Bottom line: this is the most defensible parallel route to a real gain while the current GPU job runs. It adds essentially no model parameters and little CPU pressure. However, 0.96 is a stretch-system target, not a credible single-experiment forecast.

# 1. The 0.96 feasibility test

The locked holdout is the only available numerical proxy for the public target. It does not prove that 0.96 public Recall is reachable because the public c35 oracle is unknown.

| Quantity | Value | Meaning |
|---|---|---|
| V16b public Recall | 0.921000 | Current public result [P2] |
| V16b locked-holdout Recall | 0.9228333 | Current controlled baseline [P2] |
| Locked c35 oracle Recall | 0.967667 | Maximum if the correct five can always be selected from c35 [P2, P4] |
| Public gain required | +0.039000 | 0.960000 - 0.921000 |
| Relative public error reduction | 49.37% | 0.039 / (1 - 0.921) |
| Locked-proxy gain required | +0.0371667 | 0.960000 - 0.9228333 |
| Available locked c35 headroom | +0.0448337 | 0.967667 - 0.9228333 |
| Headroom that must be captured | 82.90% | 0.0371667 / 0.0448337 |
| Error budget below the c35 oracle | 0.007667 | 0.967667 - 0.960000 |



This changes the strategy. A selector-only path to 0.96 must recover roughly 83% of all currently unused c35 headroom on the locked proxy. There is almost no room for correlated errors. Before training another ranker, compute the oracle of every permitted intervention policy.

For a policy \(S\), define:

$$R_{oracle}(S)=\frac{1}{|Q|}\sum_{q\in Q}\frac{|G_q \cap S_q^*|}{|G_q|}$$

where \(S_q^*\) is the best five-document output the policy could produce for query \(q\). Define headroom capture as:

$$Capture(S)=\frac{R_{oracle}(S)-R_{V16b}}{R_{oracle}(c35)-R_{V16b}}$$

Audit these policies first:

| Policy | Allowed change | Decision rule |
|---|---|---|
| S5 | Keep top four; replace slot 5 | Retain only if its oracle captures at least 70% of available headroom |
| S45 | Keep top three; choose two | Prefer if it captures at least 80% and remains stable across OOD splits |
| B3-12 | Keep top two; choose three from ranks 3-12 | Recommended default if it covers at least 95% of recoverable positives |
| Full-c35 | Choose any five from c35 | Use only if narrower policies cannot approach the target and full-policy validation passes |

If no policy oracle reaches 0.96 on the locked proxy, no learner under that policy can reach it. If even the c35 oracle falls below 0.96 on a relevant evaluation split, candidate generation must improve as well.

# 2. What the local evidence says

V16b is already a strong, compact multi-stage system: hybrid Stage 1, c35, a 760M Vietnamese cross-encoder, a 330M BQB component, QREL memory, engineered features, and a conservative slot-5 gate. The deployed total is approximately 1.09001B parameters [P2, P3].

The failures are informative:

- c50 degraded public Recall to 0.9181, so more candidates without better discrimination add noise [P2, P4].
- Prefix-512 cross-encoder scoring degraded public Recall to 0.9090; evidence location matters [P4].
- Generic hard-negative CE LoRA and a frozen ColBERT branch did not transfer [P4].
- V17's setwise boundary model underperformed V16b, and prior LambdaMART, listwise, and pairwise learners on old features failed [P4]. This argues against retrying a generic setwise learner with the same information.
- P0 attributed about 67.06% of recoverable headroom to errors inside c35: 64 CE-buried and 28 selector-lost cases, versus 50 pruned before c35 and 36 absent from c200 [P4]. Selection is therefore the first bottleneck, but not the only one.
- The corrected V18a audit found decisive causal evidence outside the top three in 39 of 63 audited cases; its best 1536-token pack gained +0.21363 on the targeted conditional slice and preserved headings in all audited examples [P7]. That is strong evidence-location validation, but it is not a full OOF gain estimate.

The implication is precise: do not discard E3B, and do not let it become another expensive all-c35 reranker. Use it later as one targeted feature for ambiguous candidates after a new source of candidate-specific signal has passed a CPU-only gate.

# 3. Research synthesis and transfer judgment

## 3.1 Domain features can beat larger rerankers

NOWJ's four-stage COLIEE 2026 Task 1 system compared fine-tuned 4B/8B generative rerankers with an MLP using proposition, semantic, lexical, rank, and metadata features. The feature MLP submission ranked first, while the large-generative-reranker submission ranked sixth [R1]. This is not the same dataset or metric, and those models violate this project's budget, but the transferable lesson is strong: new legal and dataset-specific signal can matter more than scaling a generic reranker.

This directly supports a tiny scorer over genuinely new candidate-conditioned features. It does not support another MLP over V16b's existing features, which local experiments already rejected.

## 3.2 Graph structure is useful, but gains are small and fragile

EACL 2023's G-DSR showed that legislative topology can improve statutory article retrieval [R5]. JuriFindIT later combined a 305M dense retriever with a roughly 0.5M-0.6M graph component; on its validation set, Recall@5 rose from 76.02 to 77.80 without reference edges and to 77.12 with them [R2]. The authors found that graph signal mostly refined plausible candidates rather than creating large recall jumps.

CaseLink achieved state-of-the-art results in its original SIGIR 2024 study and powered the second-place COLIEE 2025 Task 1 team [R3]. But a SIGIR 2025 reproducibility study could not reproduce the original CaseLink results: reproduced Recall@5 was 7.2 points lower on COLIEE 2022 and 6.6 points lower on COLIEE 2023, and heterogeneous graphs did not improve over homogeneous graphs [R4].

Therefore, V20 uses the graph as a transparent feature generator, not as a learned GNN. This keeps the experiment cheap, leak-auditable, and easy to stop.

## 3.3 Document-specific question memory is the missing complement

Document expansion by query prediction showed that retrieval improves when documents are represented by questions they can answer [R6]. Competition rules prohibit importing external generated data, so V20 does not generate anything. Instead, it reuses organizer-provided fit-fold QREL pairs as observed document-specific question prototypes.

V16b already uses QREL memory, but its route is global: current query to similar training queries to their relevant documents. V20 conditions first on the candidate:

1. Take candidate $d$ from this query's c35.
2. Collect only fit-fold training questions whose QRELs include $d$.
3. Compare the current query $q$ to that document-specific prototype set.
4. Use support-aware statistics as features for $(q,d)$.

This signal should be most valuable when the CE buries a truly relevant candidate that has historically answered semantically similar questions. Its evidence base is an adaptation of document-query expansion and V16b's existing memory mechanism; no paper validates this exact construction on DSC 2026. It must therefore pass a strict cross-fit leak audit.

### 3.4 Optimize the actual five-item decision, cautiously

Talos replaces explicit Top-K ranking operations with learned quantile thresholds and reports stable Top-K optimization under distribution shift [R7]. Its evidence is from recommender systems, not LegalIR, so V20 treats it as an objective-design hypothesis rather than an off-the-shelf answer.

The first baseline remains a regularized pointwise residual scorer. A Talos-inspired groupwise objective is tested only as an ablation. If it does not beat the simpler learner under robust OOF and OOD tests, discard it.

### 3.5 Lightweight rerankers are an escalation, not the first move

Qwen3 provides 0.6B embedding and reranking models under Apache 2.0 [R9]. Jina's 0.6B listwise v3.5 reports strong multilingual and legal-benchmark performance and up to 1.56x latency reduction, but the released weights use a non-commercial license and the evidence is a July 2026 preprint [R10]. Either would keep the full pipeline far below 4B, but both require GPU time and introduce transfer risk.

Do not add either until the CPU branch exposes residual errors that existing V16b and E3B evidence cannot separate. If a sidecar is later justified, test only one 0.6B model and count its full parameters in the end-to-end budget.

### 3.6 A vector database is infrastructure, not a relevance breakthrough

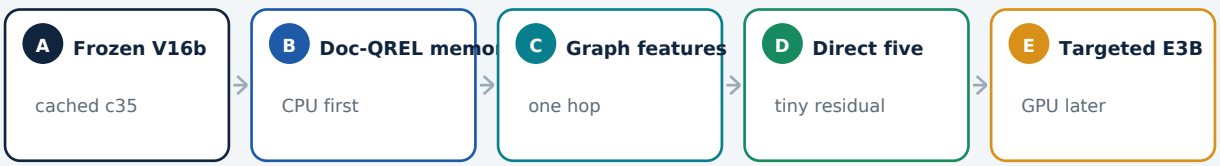
FAISS is an index for dense-vector similarity search [R11]. It can make retrieval convenient or faster; it does not improve embedding quality by itself. At this corpus size, exact inner-product search with existing BQB vectors is safer for experiments than approximate nearest-neighbor search, which adds another recall tradeoff.

Use a vector index for two bounded purposes:

- Efficient document-specific prototype lookup over training-query embeddings.
- A later multi-vector candidate complement using title, article, clause, and heading embeddings from models already in V16b.

Do not replace c35 with a new vector database until exact-search complement analysis shows unique relevant documents missing from c35.

## 4. Proposed architecture: V20 DIRECT-5




Promotion is sequential; A-D can run without loading CE/BQB. Stage E starts only after CPU gates pass.

### Stage A - Frozen V16b candidate table

Materialize one row per \((query, candidate)\): 1,197 queries x 35 candidates = 41,895 rows. Read existing V16b outputs and cached scores only. Include calibrated CE, BQB, sparse retrieval, RRF, rank, margins, QREL-memory features, candidate position, and all existing selector features.

This stage is read-only with respect to V16b. Store new outputs in an isolated experiment directory with fold, feature-version, and source hashes.

## Stage B - Candidate-conditioned prototype features

For every validation or test query, construct prototypes only from the corresponding fit fold. Use existing cached BQB query embeddings and a BM25 index over training-question text; do not load the CE.

Recommended features per candidate:

| Family | Features |
|---|---|
| Similarity | prototype top-1, top-3 mean, log-sum-exp, BM25 top-1, BQB top-1 |
| Support | QREL support count, log support, unique-template count, unique-topic count |
| Reliability | similarity x empirical-Bayes shrinkage, leave-family-out score, prototype entropy |
| Boundary | prototype margin over V16 rank-5 candidate, rank-4/5/6 gaps, disagreement with CE |
| Agreement | number of prototype channels voting candidate into top five, CE-prototype concordance |

Mandatory safeguards:

- Exclude the current query and exact normalized duplicates.
- Flag near duplicates and template siblings; report both standard and strict-template-removed OOF.
- Build every prototype index inside the fit fold; never build a global QREL map before splitting.
- Shrink low-support documents toward zero so one memorized question cannot dominate.
- Hash the fit-query IDs used by every feature row for post-hoc leak inspection.

## Stage C - Corpus-only legal graph features

Build a lightweight typed graph from organizer-provided document text and metadata. Candidate nodes can connect to law-family, article, clause, heading, explicit-citation, adjacency, and amendment/reference nodes where those relations can be extracted deterministically.

Seed the graph with frozen V16b scores and compute one-hop, degree-normalized summaries:

- Maximum and sum of neighboring candidate scores by relation type.
- Number of independent high-ranked neighbors supporting a candidate.
- Candidate degree and inverse-degree-weighted support.
- Agreement between document hierarchy and the query's extracted statute mentions.
- Graph support margin against the current fifth candidate.

No label propagation crosses folds. No test QREL edge enters the graph. Do not expand c35 in the first graph experiment; score only candidates already present. Add one relation type at a time so each contribution is attributable.

## Stage D - Direct five-item decision

Train a tiny residual scorer:

$$s'(q,d)=s_{V16}(q,d)+clip(r_\theta(x_{q,d}),-\delta_q,+\delta_q)$$

where $r_\theta$ is L2 logistic regression or a two-layer MLP with fewer than one million parameters. Set $\delta_q$ from the local rank-margin scale so the model cannot arbitrarily rewrite confident V16b orderings.

Compare three objectives on identical folds and features:

1 BCE residual: the stable reference.

2 Boundary pair residual: positives versus candidates around the fifth-score boundary; use semi-hard, not maximally hard, negatives.

3 Top-5 quantile residual: a Talos-inspired surrogate centered on the query's learned fifth-score threshold.

Compare all three policy envelopes S5, S45, and B3-12. Full-c35 selection is a gated fourth envelope, not the default.

### Stage E - Targeted E3B evidence escalation

After the CPU branch passes, add only the single best validated 1536-token E3B evidence score for the ambiguous band. Do not score all six evidence configurations over all c35 candidates.

For 1,197 queries:

| Scoring plan | Pair evaluations | Relative work |
|---|---|---|
| Six configurations over c35 | 251,370 | 100.0% |
| One configuration over 12 candidates | 14,364 | 5.71% |
| Reduction | 237,006 fewer | 94.29% |

Measure 100 pairs first, record throughput and VRAM, and extrapolate rather than guessing runtime. Cache by query ID, candidate ID, evidence-config hash, tokenizer hash, and model hash.

## 5. Compute and parameter budget

### What can run now while the machine is busy

The first four experiments are CPU-first and should not load either neural model:

- Build the 41,895-row candidate table from cached files.
- Build fit-fold BM25 prototype indices and compute cached-BQB similarities.
- Parse deterministic corpus relations and compute sparse one-hop graph features.
- Fit regularized logistic/MLP residuals and run paired bootstrap evaluation.

Limit BLAS/OpenMP to two or four threads, lower process priority if available, and use Parquet or memory-mapped arrays. A 41,895-row table with a few hundred numeric features is small; the dominant cost should be feature joins and resampling, not model inference. This branch can run concurrently with a GPU-heavy job if it does not reload BQB or CE weights.

GPU work begins only at Stage E. Schedule it after the current V18 scoring process releases VRAM, or run a very small throughput pilot only if there is guaranteed headroom.

## Deployed parameter accounting

| Component | Parameters | Status |
|---|---|---|
| Existing V16b end-to-end pipeline | 1.09001B | Frozen baseline [P2, P3] |
| DIRECT-5 residual selector | <0.001B | Proposed; CPU |
| Prototype index and graph features | 0 | Proposed; non-parametric |
| E3B evidence packing/scoring | 0 additional | Reuses existing CE |
| Core V20 total | <1.09101B | Safely below 4B |
| Optional one 0.6B sidecar | +0.6B | Later only |
| V20 plus sidecar | <1.69101B | Still safely below 4B |

Quantization lowers memory, not parameter count. Always report full model parameters for competition compliance.

# 6. Experiment program and promotion gates

## E0 - Selection ceiling audit

Hypothesis: most usable headroom is accessible from a narrower ambiguity band than full c35.

Setup: reproduce V16b OOF, robust OOF, locked-holdout score, per-query c35 contents, gold ranks, and policy oracles S5, S45, B3-12, and Full-c35.

Outputs: policy oracle Recall, headroom capture, recoverable-positive coverage, number of positions changed, and candidate-rank histograms.

Gate: no learner is trained for a policy whose oracle cannot beat V16b by at least +0.005. A policy intended for the 0.96 roadmap must itself have oracle Recall at or above 0.96 on the locked proxy.

## E1 - Candidate-conditioned prototype audit

Hypothesis: relevant documents buried by the CE have stronger document-specific question-prototype support than competing rank-4-to-12 candidates.

Variants: BM25 prototypes; cached-BQB prototypes; calibrated hybrid; hybrid with support shrinkage; strict duplicate/template-removed hybrid.

Metrics: boundary AUPRC, positive mean reciprocal rank within the policy band, unique correct rescues, false promotion count, Recall@5 after a one-feature monotonic rerank, and per-fold deltas.

Gate: proceed only if at least one leak-safe variant supplies independent signal: either boundary AUPRC improves by at least 0.03 over existing V16 features or a one-feature policy simulation exposes at least +0.004 oracle-addressable OOF headroom.

## E2 - Tiny residual selector

Hypothesis: new prototype features allow a simple, strongly regularized model to beat V16b without repeating V17's setwise overfit.

Models: L2 logistic residual; monotonic generalized additive residual; two-layer MLP only if both simpler models pass.

Objectives: BCE, boundary semi-hard pair, and Top-5 quantile surrogate.

Gate to retain a model:

- Standard OOF Recall gain >= +0.003.

- Robust OOF Recall gain >= +0.002.
- Paired query-bootstrap 95% lower bound > 0.
- Positive delta on at least three of the four existing OOD partitions.
- Worst OOD delta > -0.001.
- Precision loss <= 0.003 and intervention precision >= 0.75.

These are recommended decision thresholds, not facts from the literature. Freeze them before viewing locked-holdout results.

### E3 - Graph feature ablation

Hypothesis: deterministic legislative relations add signal that prototypes and CE scores miss.

Ablations: hierarchy only; explicit references only; same-law/adjacency only; all relations; all relations without high-degree nodes; one-hop versus two-hop.

Gate: the graph must add at least +0.001 standard OOF Recall over the prototype selector, remain non-negative under strict-template OOF, and be positive on at least three OOD partitions. Otherwise remove it. Do not train a GNN merely because static graph features fail.

### E4 - Targeted E3B fusion

Hypothesis: decisive causal evidence disambiguates residual errors only after candidate-specific memory narrows the boundary set.

Setup: score one evidence configuration over the selected band, add raw score, calibrated margin, evidence position, and agreement features; compare selector with and without E3B.

Gate: E3B must add >= +0.002 robust OOF Recall over the CPU branch and retain a positive bootstrap lower bound. If it helps only the 63-case audit slice, do not promote it.

### E5 - Candidate complement, only if required

Hypothesis: a multi-vector exact-search complement can recover documents absent from c35 without recreating c50 noise.

Setup: use existing BQB to embed title, article, clause, heading, and whole-document views; exact-search each view; fuse with RRF; separately test one-hop graph expansion. Add only candidates not already in c35.

Gate: evaluate unique gold rescue before any reranking. Require at least +0.004 candidate-oracle Recall and no reduction in original-c35 coverage. If the complement only reshuffles existing c35 documents, stop.

## 7. Evaluation protocol

Every reported number should include:

- Standard five-fold OOF and the existing robust OOF protocol.
- The four existing OOD/bootstrap partitions from the boundary research.
- Per-query paired delta, 10,000-sample paired bootstrap confidence interval, and fold-level sign consistency.
- Recall as primary; Precision, F1, intervention precision, and false replacement count as guardrails.
- Oracle Recall for the active candidate pool and active decision policy.
- Breakdown by original V16 rank of rescued positives and demoted positives.

- Duplicate/template leakage audit and fit-query provenance hashes.
- Wall time, peak RAM, peak VRAM, pair count, and deployed parameter count.

Run the locked holdout once, only after model, features, thresholds, policy, and seeds are frozen. A public submission should follow only if the locked result clears the same predeclared gain and guardrail logic.

# 8. Direction comparison

| Direction | New signal | Cost now | Main risk | Decision |
|---|---|---|---|---|
| Candidate-conditioned QREL prototypes | High and specific | Low CPU | Fold leakage / low-support memorization | Start now |
| Direct top-five residual scorer | Aligns decision with max-five metric | Low CPU | Small-data overfit | Start after E0/E1 |
| Static legal graph features | Complementary structure | Low-moderate CPU | Noisy edges; fragile transfer | Ablate after prototypes |
| Targeted V18 E3B | Better causal evidence | Moderate GPU later | Audit-slice gain may not generalize | Keep, narrow, and gate |
| Full vector database replacement | Mostly infrastructure | Moderate | No new relevance signal | Do not replace c35 |
| Learned GNN / CaseLink clone | Higher-order structure | High GPU/engineering | Reproducibility and transfer | Do not start now |
| 0.6B listwise reranker | Cross-document comparison | GPU later | Domain/license/transfer risk | Optional escalation |
| 4B+ or multi-agent LegalRAG | Potential reasoning | Disallowed | Parameter and data compliance | Exclude |

# 9. Conditional roadmap to 0.96

These are go/no-go milestones, not forecasts:

| Milestone | Required evidence | Next action |
|---|---|---|
| M0: Policy feasibility | Policy oracle and c35 oracle support target | Choose S5, S45, B3-12, or Full-c35 |
| M1: CPU breakthrough | >= +0.003 standard and +0.002 robust OOF | Freeze prototype selector |
| M2: Evidence gain | Targeted E3B adds >= +0.002 robust OOF | Fuse one evidence config |
| M3: Candidate gap | Residual errors are outside c35 | Add exact multi-vector complement |
| M4: Near-oracle stack | Independent error sets; lower CI > 0 at each layer | Consider one 0.6B sidecar |
| M5: Submission | Locked holdout clears frozen gates | Submit once; do not chase public noise |

A plausible successful first cycle is movement from ~0.9228 into the high 0.92s. Reaching 0.96 would then require repeated, independently validated gains and near-oracle selection. If the first CPU branch fails its gates, the correct conclusion is not to add a larger selector; it is that V16b lacks the necessary independent information and effort should move to candidate/evidence representation.

# 10. Immediate implementation queue

1. P0-D Policy audit (CPU): compute policy oracles and select the smallest intervention envelope capable of the target.
2. P1-D Prototype table (CPU): build cross-fit candidate-conditioned BM25/BQB features with provenance hashes.
3. P2-D Leak and signal audit (CPU): run duplicate/template removal and boundary separation diagnostics before training.
4. P3-D Direct-five learners (CPU): BCE residual first; then boundary-pair and Top-5-quantile ablations.
5. P4-D Graph features (CPU): add one deterministic relation family at a time; stop at the first failed gate.
6. P5-D Targeted E3B (GPU later): score one 1536-token configuration on the chosen ambiguity band.
7. P6-D Frozen fusion: one robust OOF selection, one locked-holdout evaluation, then a submission decision.

Parallel-run safeguards: read V16b caches only; write a new experiment namespace; cap CPU threads; do not instantiate CE/BQB models; make all fold and feature hashes explicit; avoid simultaneous disk-heavy joins during V18 checkpoint writes.

# 11. Evidence-gap matrix

| Claim or component | Direct evidence | Transfer gap | How V20 closes the gap |
|---|---|---|---|
| Domain features can beat large rerankers | NOWJ COLIEE 2026 [R1] | Different case-law task and metric | Test only on frozen local OOF; require robust gains |
| Legislative graphs add complementary signal | G-DSR and JuriFindIT [R2, R5] | Different languages/corpora; modest gains | Use static one-hop features with relation ablations |
| Full graph systems are risky | CaseLink reproduction [R4] | Case law rather than Vietnamese statute retrieval | Avoid GNN; enforce OOD and bootstrap gates |
| Questions that a document answers are useful representations | doc2query [R6] | Generated queries; not candidate-conditioned QREL memory | Use organizer QREL questions only and strict cross-fit |
| Top-K aligned loss may improve selection | Talos [R7] | Recommender systems, not LegalIR | Treat as one ablation against BCE residual |
| Semi-hard negatives are preferable in Vietnamese LegalIR | ICCCI 2025 study [R8] | Different dataset and retriever | Use only at boundary-pair stage; compare to BCE |
| 0.6B rerankers can be strong | Qwen3 and Jina v3.5 [R9, R10] | General benchmarks; license and transfer uncertainty | Optional sidecar only after residual-error audit |
| Vector database improves relevance | None | Indexing is not representation learning | Use exact indexing as infrastructure only |
| DIRECT-5 can reach 0.96 | None | Public oracle unknown; requires 82.9% headroom capture | Policy oracle first, then staged lower-bound gates |

# 12. Claim-to-source ledger

| ID | Report claim | Source support | Confidence |
|---|---|---|---|
| C1 | V16b public 0.9210, holdout 0.9228333, ~1.09001B params | [P2, P3] | High; project record |
| C2 | Locked c35 oracle is 0.967667 | [P2, P4] | High; project record |
| C3 | Much local headroom is inside c35 | [P4] | High for audited local split |
| C4 | V18a found useful evidence outside top three | [P7] | High for audited slice; not full OOF |
| C5 | Feature MLP beat large rerankers in COLIEE 2026 | [R1] | High for that competition |
| C6 | Legal graph signal can improve retrieval | [R2, R3, R5] | Medium-high; multiple domains |
| C7 | CaseLink gains are not reliably reproducible | [R4] | High for reproduced datasets |
| C8 | Document-query representations can improve retrieval | [R6] | High generally; medium transfer |
| C9 | Quantile loss can target Top-K accuracy | [R7] | High in recommender setting; low-medium transfer |
| C10 | 0.6B rerankers are viable under the budget | [R9, R10] | High for parameter count; medium for task gain |
| C11 | A vector index does not itself improve embeddings | [R11] plus system inference | High |
| C12 | DIRECT-5 is the best next branch | Synthesis of [P2-P7, R1-R11] | Medium; experiment required |

# 13. Sources

## Project sources

- [P1] `README(1).md` and `README(2).md` - competition rules, execution, and parameter budget.
- [P2] `CURRENT_STATE(1).md` - V16b scores, pipeline status, and locked evaluation.
- [P3] `V16_RGM_DISCLOSURE(1).md` - QREL memory, features, and slot-5 gate.
- [P4] `V16_BOUNDARY_RESEARCH_DISCLOSURE.md` - P0-P4 findings, OOD/bootstrap analysis, local failures.
- [P5] `FINAL_V11H_END_TO_END_SUMMARY(2).md` - foundational Stage-1 to c35 to CE to selector workflow.
- [P6] `DSC2026_Task1_LegalIR_Data_Overview(1).pdf` - task and dataset overview.
- [P7] `Pasted markdown(2).md` - corrected V18a evidence audit and V18b status.

## External primary research and official documentation


- [R1] NOWJ@COLIEE 2026: Adaptive Pipelines for Legal Retrieval and Reasoning. source link
- [R2] JuriFindIT: an Italian legal retrieval dataset. Findings of EACL 2026. source link
- [R3] CaseLink: Inductive Graph Learning for Legal Case Retrieval. SIGIR 2024. source link
- [R4] A Reproducibility Study of Graph-Based Legal Case Retrieval. SIGIR 2025. source link
- [R5] Finding the Law: Enhancing Statutory Article Retrieval via Graph Neural Networks. EACL 2023. source link
- [R6] Document Expansion by Query Prediction. arXiv 2019. source link
- [R7] Talos: Optimizing Top-K Accuracy in Recommender Systems. WWW 2026. source link
- [R8] Optimizing Legal Document Retrieval in Vietnamese with Semi-Hard Negative Mining. ICCCI 2025. source link
- [R9] Qwen3 Embedding: Advancing Text Embedding and Reranking Through Foundation Models. 2025. source link
- [R10] jina-reranker-v3.5: An Efficient Listwise Reranker with Hybrid Attention and Self-Distillation. 2026. source link
- [R11] FAISS official documentation. source link


# Final recommendation

Run V20 DIRECT-5 beside V18 now. The order is: policy oracle -> candidate-conditioned prototypes -> tiny direct-five residual -> static graph ablation -> targeted E3B. Do not launch a vector-database replacement, a GNN, or a new 0.6B reranker until the earlier gates prove that the missing signal exists and transfers.

This path preserves V16b, consumes negligible additional parameters, avoids competing for GPU during the current run, and directly attacks the c35-to-five bottleneck the local evidence identifies. It is the best available route to a measurable breakthrough, while keeping the 0.96 target honest: first prove reachability, then earn each layer of headroom.